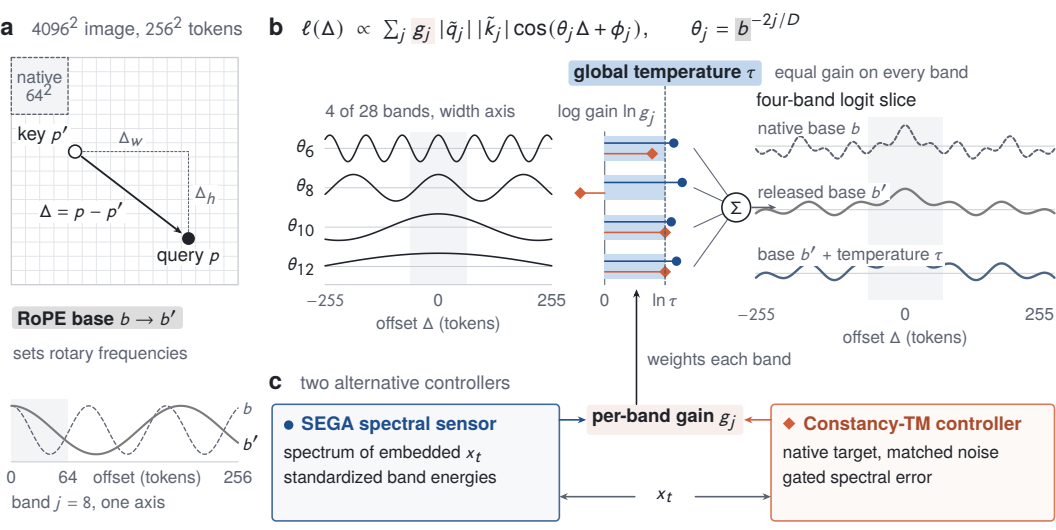
a
4096² image, 256² tokens
native 64²
key p′
Δw
Δh
Δ = p − p′
query p
RoPE base b → b′
sets rotary frequencies
b
b′
0
64
offset (tokens)
256
band j = 8, one axis
b
ℓ(Δ) ∝ ∑j gj |q̃j| |k̃j| cos(θjΔ + ϕj),
θj = b^−2j/D
global temperature τ
equal gain on every band
4 of 28 bands, width axis
log gain ln gj
four-band logit slice
θ6
θ8
θ10
θ12
native base b
released base b′
base b′ + temperature τ
Σ
−255
0
255
offset Δ (tokens)
ln τ
weights each band
c
two alternative controllers
SEGA spectral sensor
spectrum of embedded xt
standardized band energies
per-band gain gj
Constancy-TM controller
native target, matched noise
gated spectral error
xt